

Baseline Verification — Synthesis Report

4DVarNet-FM: Baseline Verification Report

Common parameters:

T_max = 3.0s dt = 0.01 Steps = 300
obs_interval = 20 R_var = 0.5 B_var = 2.0

DA Baseline parameters:

DA window steps (DWS) = 300
Weak-4DVar : opt_steps=150, lr=0.02
Strong-4DVar: max_iter=40, lr=0.1
EnKF : N_ensemble=30, inflation=1.2
ETKF : N_ensemble=30, inflation=1.0 (deterministic transform)

Case studies:

Parameter	CS1	CS2
Forcing	Noise-free (true W_L)	Corrupted (OU process)
Parameter bias	0%	15%
Forcing-state bias	0.0	0.15
Coupling	Linear	Quartic

Test data:

200 trajectories per case study, generated with seed=123 (CS1) / 124 (CS2)
Forcing corrupted by Ornstein-Uhlenbeck noise (tau_eta=5.0, sigma_eta=0.707)